

Application of Machine Learning to Predict Building Damage in the Nepal Earthquake

Machine Learning Mini-Project | Random Forest Classifier

1. Problem Statement

Estimating damage across large numbers of earthquake-affected buildings through individual structural assessments can be time-consuming. This project applies supervised machine learning to predict an observed building damage category from the structural, geographic, and usage attributes recorded after the 2015 Nepal earthquake. The intended outcome is an automated classification demonstration, not a substitute for professional safety inspection.

2. Dataset

The project uses the publicly available *Richter's Predictor: Modeling Earthquake Damage* training data, associated with the 2015 Nepal earthquake. **train_values.csv** provides 260,601 building records and 38 predictive attributes; **train_labels.csv** provides the matching damage grades. The files are joined by **building_id**. No missing entries were found in the uploaded training data.

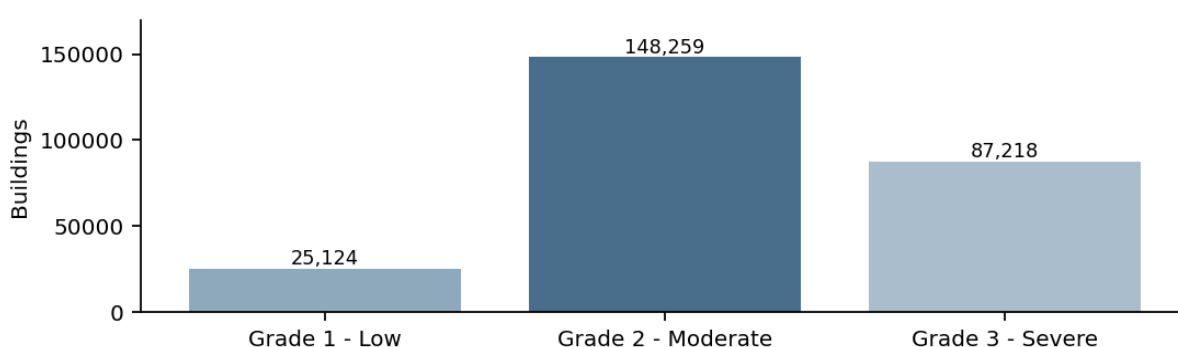


Figure 1. Class distribution in the labeled data (260,601 buildings).

Damage levels are **1: low damage**, **2: moderate damage**, and **3: severe / near-complete destruction**. Grade 2 is the majority category (148,259 records), followed by Grade 3 (87,218) and Grade 1 (25,124).

3. Methodology and Implementation

The two CSV files are merged; the identifier **building_id** is excluded from training. Categorical attributes are transformed using one-hot encoding, yielding **68 encoded input features**. The data is split into **80% training (208,480)** and **20% held-out testing (52,121)**, stratified by damage grade with `random_state=42`. A **RandomForestClassifier** with **100 trees**, `random_state=42`, and `n_jobs=-1` is trained using scikit-learn. Python, pandas, NumPy, Matplotlib, seaborn, and Google Colab are used for execution and visualization.

Random Forest combines predictions from many decision trees through majority voting. One algorithm is used throughout this implementation. Model evaluation is performed on the held-out test records, rather than on training records.

Results, Discussion and Conclusion

4. Evaluation Results

The original Random Forest achieved **71.60% test accuracy** on 52,121 held-out buildings. Since this is single-label multiclass classification, the micro-averaged F1-score equals the accuracy (approximately 0.716). Training took **73.01 seconds** in the reported Colab run. The class-wise results are summarized below.

Damage class	Precision	Recall	F1-score	Test records
1 - Low	0.64	0.48	0.55	5,025
2 - Moderate	0.72	0.82	0.77	29,652
3 - Severe	0.72	0.60	0.65	17,444

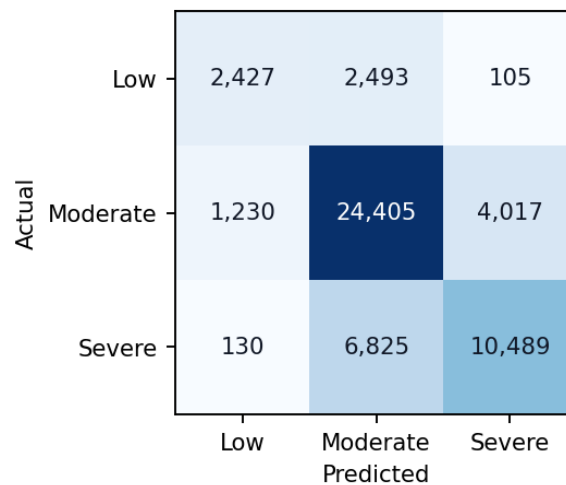


Figure 2. Confusion matrix on the held-out test set (rows: actual, columns: predicted).

The model correctly classified **37,321** buildings and misclassified **14,800**. Moderate damage is the easiest category to recognize (recall 0.82). A major limitation is that **6,825** severely damaged buildings were classified as moderate damage, giving severe-damage recall of just 0.60. The most prominent features in the model's importance chart were geographic identifiers, building age, area, and height. Importance indicates predictive use, not causality.

5. Prediction Demonstration

A held-out example with geographic level-1 ID 18, two floors, age 5, area percentage 8, and height percentage 5 was predicted as **Grade 3 (severe damage)**, matching its recorded Grade 3 label. The estimated model probabilities were 1.00% low, 37.85% moderate, and 61.15% severe. These probabilities are not guaranteed confidence measures.

6. Conclusion

A single Random Forest model successfully produced reproducible damage-grade predictions and achieved 71.60% held-out test accuracy. Its strengths include a straightforward training pipeline and interpretable evaluation outputs. Limitations include uneven class performance, especially under-identification of severe damage, possible dependence on geographic identifiers, and the fact that the data describes a historical earthquake rather than future events. The prototype is suitable as an academic demonstration, not an operational safety-assessment tool.

References

- [1] DrivenData, *Richter's Predictor: Modeling Earthquake Damage*. <https://www.drivendata.org/competitions/57/nepal-earthquake/>
- [2] Yitian Liang, *Application of Machine Learning Methods to Predict the Level of Buildings Damage in the Nepal Earthquake*, Stanford CS229 Final Project Report, 2021. This project uses only the Random Forest method discussed in the reference.